WEYL–CHAMBER GEOMETRY, POINCARÉ INEQUALITY MEETS SCHATTEN CLASS PROPERTIES OF DUNKL–RIESZ TRANSFORM COMMUTATORS

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Abstract.

Contents

1. Introduction

Definition 1.1. Suppose $1 < p < \infty$. We say that a function $f \in L^p_{\mathrm{loc}}(\mathbb{R}^N, dw)$ belongs to Dunkl Besov space $B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)$ if

$$\|f\|_{B^p_{\mathrm{Dunkl}}(\mathbb{R}^N)} := \left(\int_{\mathbb{R}^N}\int_{\mathbb{R}^N}\left(\frac{d(x,y)}{\|x-y\|}\right)^p \frac{|f(x)-f(y)|^p}{w(B(x,d(x,y)))^2}dw(x)dw(y)\right)^{1/p} < +\infty.$$

1.1. Notation. Conventionally, we let $\mathbb{N}$ denote the set of positive integers and set $\mathbb{Z}_+ := \{0\} \cup \mathbb{N}$. Throughout the paper, the symbols C and c denote positive constants (possibly varying from line to line) that may depend only on N, p, and the structural data (R, k) of the Dunkl setting, but are independent of the main parameters (e.g., the symbol b and auxiliary decay factors). We write $A \lesssim B$ to mean that $A \le CB$ for some constant $C > 0$, and $A \sim B$ to mean that both $A \lesssim B$ and $B \lesssim A$ hold. For $\lambda > 0$ and a cube Q with sidelength $\ell(Q)$, write λQ for the cube concentric with Q and sidelength $\lambda\,\ell(Q)$. Similarly, for every ball $B = B(x, r)$, set $\lambda B := B(x, \lambda r)$. For any $1 \le p \le \infty$, we denote by p' the conjugate of p, which satisfies $\frac{1}{p} + \frac{1}{p'} = 1$. We denote the average of a function f over a μ-measurable set E by

$$(f)_E := \fint_E f(x)d\mu(x) := \frac{1}{\mu(E)}\int_E f(x)d\mu(x). \tag{1.1}$$

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For a subset $E \subset \mathbb{R}^N$, we denote by χ_E its characteristic function.

Denote by $\{e_j\}_{j=1}^N$ the standard orthonormal basis on $\mathbb{R}^N$.

2. Preliminaries

2.1. Harmonic analysis in the Dunkl setting. Let $\mathbb{R}^N$ be the Euclidean space equipped with the usual scalar product $\langle \cdot, \cdot \rangle$ and the induced norm $\| \cdot \|$. For a nonzero vector $v \in \mathbb{R}^N$, the reflection σ_v with respect to the hyperplane $v^\perp$ orthogonal to v is defined by

$$\sigma_v x := x - 2\frac{\langle x, v\rangle}{\|v\|^2} v.$$

A finite set $R \subset \mathbb{R}^N \backslash \{0\}$ is called a root system if $\sigma_v(R) = R$ for every $v \in R$. A root system R is called a normalized reduced root system if $\|v\|^2 = 2$ for every $v \in R$. The finite group G generated by the reflections $\{\sigma_v\}_{v\in R}$ is called the Weyl group (reflection group) of the root system R. Let $k : R \to [0, \infty)$ be a multiplicity function, which is a G-invariant function fixed throughout the paper. We will denote by $\mathcal{O}(x) = \{\sigma(x) : \sigma \in G\}$ the G-orbit of a point $x \in \mathbb{R}^N$, and $\mathcal{O}(E) = \bigcup_{x\in E} \mathcal{O}(x)$ the G-orbit of $E \subset \mathbb{R}^N$.

Define the Dunkl measure dw on $\mathbb{R}^N$ by

$$dw(x) := \prod_{v\in R} |\langle x, v\rangle|^{k(v)} dx,$$

where dx stands for the Lebesgue measure on $\mathbb{R}^N$. Let $B(x, r) := \{y \in \mathbb{R}^N : \|x - y\| < r\}$ be the Euclidean ball centred at x with radius $r > 0$. Let $\mathbf{N} := N + \sum_{v\in R} k(v)$. Note that

$$w(B(tx, tr)) = t^{\mathbf{N}} w(B(x, r)), \text{ for } x \in \mathbb{R}^N, \ t, r > 0.$$

Furthermore,

$$w(B(x, r)) \sim r^N \prod_{v\in R} (|\langle x, v\rangle| + r)^{k(v)}, \tag{2.1}$$

where the implicit constant only depends on N, R and k. It follows from (2.1) that there exists a constant $C \geq 1$ such that for every $x \in \mathbb{R}^N$,

$$C^{-1} \left(\frac{r_2}{r_1}\right)^N \leq \frac{w(B(x, r_2))}{w(B(x, r_1))} \leq C \left(\frac{r_2}{r_1}\right)^{\mathbf{N}}, \text{ for } 0 < r_1 < r_2, \tag{2.2}$$

so the numbers $\mathbf{N}$ and N are called the upper and lower homogeneous dimensions, respectively.

The Dunkl operators T_ξ are defined by

$$T_\xi f(x) := \partial_\xi f(x) + \sum_{v\in R} \frac{k(v)}{2} \langle v, \xi\rangle \frac{f(x) - f(\sigma_v(x))}{\langle v, x\rangle},$$

where $\partial_\xi f$ denotes the directional derivative of f in the direction ξ. Furthermore, the Dunkl Laplacian is defined by

$$\Delta_{\text{Dunkl}} f(x) := \sum_{j=1}^N T_{e_j}^2 f(x) = \Delta_{\mathbb{R}^N} f(x) + \sum_{v\in R} k(v) \delta_v f(x),$$

where $\Delta_{\mathbb{R}^N}$ is the Laplacian on $\mathbb{R}^N$ and δ_v is defined by

$$\delta_v f(x) := \frac{\partial_v f(x)}{\langle v, x\rangle} - \frac{f(x) - f(\sigma_v(x))}{\langle v, x\rangle^2}.$$

In this article, we consider the Dunkl-Riesz transforms defined by $R_{\text{Dunkl},j} := T_{e_j}(-\Delta_{\text{Dunkl}})^{-1/2}$, $j = 1, 2, \ldots, N$.

Definition 2.1. The Weyl chambers are defined as the closures of connected components of the set

$$\left\{x \in \mathbb{R}^N : \langle x, v\rangle \neq 0 \quad for\ all\ v \in R\right\}. \tag{2.3}$$

Define $d(x,y) := \min_{\sigma\in G} \|x - \sigma(y)\|$ as the distance between two G-orbits $\mathcal{O}(x)$ and $\mathcal{O}(y)$. Recall from [7, Chapter VII, proof of Theorem 2.12] that

$$d(x,y) = \|x - y\| \text{ if and only if } x, y \in \Gamma \tag{2.4}$$

for some closed Weyl chamber Γ.

It is clear that $\mathcal{O}(B(x,r)) = \{y \in \mathbb{R}^N : d(x,y) < r\}$. Moreover,

$$w(B(x,r)) \leq w(\mathcal{O}(B(x,r))) \leq |G|w(B(x,r))$$

for all $x \in \mathbb{R}^N$ and $r > 0$. We also denote $B^d(x,r) := \{y \in \mathbb{R}^N : d(x,y) < r\}$, i.e., $B^d(x,r) = \mathcal{O}(B(x,r))$.

Denote by $K_{\text{Dunkl},j}(x,y)$ the integral kernel of $R_{\text{Dunkl},j}$. We recall the kernel estimates for the Riesz transforms in the Dunkl setting proved in [5]. Although these estimates may not be sharp (see the comment following Theorem 2.1 in [5, p. 10]), they suffice for our purposes.

Lemma 2.2. There exists a constant C such that for $j \in \{1, 2, \ldots, N\}$ and every x, y with $d(x,y) \neq 0$,

$$|K_{\text{Dunkl},j}(x,y)| \leq C\frac{d(x,y)}{\|x-y\|}\frac{1}{w(B(x,d(x,y)))}.$$

Proof. The proof is given in [5, Theorem 1.1]. □

2.2. Systems of dyadic cubes.

Definition 2.3. Let $X \subseteq \mathbb{R}^N$ be a nonempty open set. A countable family $\mathscr{D} := \bigcup_{k\in\mathbb{Z}} \mathscr{D}_k$, $\mathscr{D}_k := \{Q^k_\alpha : \alpha \in \mathscr{A}_k\}$, of Borel sets $Q^k_\alpha \subseteq X$ is called a system of dyadic cubes over X if it has the following properties:

(i) For each $k \in \mathbb{Z}$, we have

$$X = \bigcup_{\alpha\in\mathscr{A}_k} Q^k_\alpha;$$

(ii) If $\ell \geq k$, then either $Q^\ell_\beta \subseteq Q^k_\alpha$ or $Q^k_\alpha \cap Q^\ell_\beta = \varnothing$;

(iii) For each (k, α) and each $\ell \leq k$, there exists a unique β such that $Q^k_\alpha \subseteq Q^\ell_\beta$;

(iv) For each (k,α) there exist 2^N indices β such that
$$Q_\beta^{k+1} \subseteq Q_\alpha^k, \text{ and } Q_\alpha^k = \bigcup_{\substack{Q\in\mathscr{D}_{k+1}\\ Q\subseteq Q_\alpha^k}} Q;$$

(v) For each (k,α), one has
$$B_X(x_\alpha^k, A_1 2^{-k}) \subseteq Q_\alpha^k \subseteq B_X(x_\alpha^k, A_2 2^{-k}) =: B(Q_\alpha^k)$$
for some positive constants A_1, A_2 depending only on the geometry of X, where we set $B_X(x,r) := B(x,r)\cap X$ for all $x\in X$ and $r>0$.

(vi) If $\ell \geq k$ and $Q_\beta^\ell \subseteq Q_\alpha^k$, then
$$B(Q_\beta^\ell) \subseteq B(Q_\alpha^k).$$

Each Q_α^k is called a dyadic cube of generation k, with center $x_\alpha^k \in Q_\alpha^k$ and side-length $\ell(Q_\alpha^k) = 2^{-k}$.

Notation 2.4. For every dyadic cube $Q\in\mathscr{D}$, we denote by c_Q and $\ell(Q)$ its center and side-length, respectively.

Notation 2.5. Let
$$\mathbb{D}^0 := \bigcup_{k\in\mathbb{Z}} \mathbb{D}_k^0$$
be the standard system of dyadic cubes over $\mathbb{R}^N$. Here each dyadic cube is a product of half-open intervals
$$[m_1 2^{-k}, (m_1+1)2^{-k}) \times\cdots\times [m_N 2^{-k}, (m_N+1)2^{-k}),$$
with $(m_1,\ldots,m_N)\in\mathbb{Z}^N$. Moreover, we let
$$\mathbb{D}_\Gamma := \bigcup_{k\in\mathbb{Z}} \mathbb{D}_{\Gamma,k}$$
be the system of dyadic cubes over a fixed chamber Γ, whose construction will be given in Corollary 3.7.

Lemma 2.6. There exist 2^N systems of dyadic cubes $\mathbb{D}^1, \mathbb{D}^2, \ldots, \mathbb{D}^{2^N}$ such that for any ball $B = B(x_B, r_B) \subset \mathbb{R}^N$ with $\frac{2^{-k-2}}{3} \leq r_B < \frac{2^{-k-1}}{3}$ for some $k\in\mathbb{Z}$, there exists a cube $Q \in \bigcup_{\nu=1}^{2^N} \mathbb{D}_k^\nu$ such that
$$B\subset Q\subset 12\sqrt{N}B.$$

Proof. • Step 1. Recall the one-dimensional setting given in [10, Section 2].

Let $\mathbb{D}^{(0)}(\mathbb{R})$ be the standard system of dyadic intervals over $\mathbb{R}$ (see Notation 2.5). Let $\delta = 1/3$ and define
$$\mathbb{D}^{(\delta)}(\mathbb{R}) = \bigcup_{k\in\mathbb{Z}} \mathbb{D}_k^{(\delta)}(\mathbb{R}),$$
where for $k\geq 0$,
$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \{I+\delta \mid I\in\mathbb{D}_k^{(0)}(\mathbb{R})\},$$

while for $k<0$ and k even,

$$\mathbb{D}_k^{(\delta)}(\mathbb{R}) = \Big\{\Big[\frac{m}{2^k}+\delta+\sum_{j=(k/2)+1}^{0}2^{-2j},\ \frac{m+1}{2^k}+\delta+\sum_{j=(k/2)+1}^{0}2^{-2j}\Big)\ \Big|\ m\in\mathbb{Z}\Big\}.$$

These choices together with the nestedness property property completely determine the collections $\mathbb{D}_k^{(\delta)}(\mathbb{R})$ for $k<0$, k odd.

Fix an interval $J\subset\mathbb{R}$ with $2^{-k-1}/3\le |J|<2^{-k}/3$ for some $k\in\mathbb{Z}$. Now we set $A_k=\{m\cdot 2^{-k}\mid m\in\mathbb{Z}\}$ for every fixed k and

(1) $A_k^{(\delta)}=\{\delta+m\cdot 2^{-k}\mid m\in\mathbb{Z}\}$ for $k\ge 0$,
(2) $A_k^{(\delta)}=\{\delta+\sum_{j=(k/2)+1}^{0}2^{-2j}+m\cdot 2^{-k}\mid m\in\mathbb{Z}\}$ for $k<0$, k even, and
(3) $A_k^{(\delta)}=\{\delta+\sum_{j=(k-1)/2+1}^{0}2^{-2j}+m\cdot 2^{-k}\mid m\in\mathbb{Z}\}$ for $k<0$, k odd.

Note that for any two different points $a,b\in A_k\cup A_k^{(\delta)}$, we have $|a-b|\ge 2^{-k}/3>|J|$. Thus, there is at most one element of $A_k\cup A_k^{(\delta)}$ contained in J. Hence, $J\cap A_k=\emptyset$ or $J\cap A_k^{(\delta)}=\emptyset$. Therefore, J must be contained in some dyadic interval $I\in\mathbb{D}_k^{(0)}$ or $I\in\mathbb{D}_k^{(\delta)}$ and $|I|=2^{-k}\le 6|J|$.

• Step 2. We extend the construction to $\mathbb{R}^N$ $(N\ge 1)$.

For $\vartheta=(\vartheta_1,\vartheta_2,\dots,\vartheta_N)\in\{0,\delta\}^N$, define

$$\mathbb{D}^{(\vartheta)} := \bigcup_{k\in\mathbb{Z}}\mathbb{D}_k^{(\vartheta)},\quad \mathbb{D}_k^{(\vartheta)} := \left\{\prod_{j=1}^{N} I_j : I_j\in\mathbb{D}_k^{(\vartheta_j)}\right\}.$$

Applying Step 1 to $\pi_i(B)$, where $\pi_i(B)$ denotes the i-th coordinate projection, we obtain $I_i\in\mathbb{D}_k^{(0)}$ or $I_i\in\mathbb{D}_k^{(\delta)}$ with $\pi_i(B)\subset I_i$ and $|I_i|=2^{-k}\le 12r_B$. Let $\vartheta_i\in\{0,\delta\}$ be the corresponding shift so that $I_i\in\mathbb{D}_k^{(\vartheta_i)}$, set $\vartheta:=(\vartheta_1,\dots,\vartheta_N)$, and let $Q=\prod_{j=1}^{N}I_j\in\mathbb{D}_k^{(\vartheta)}$. It follows that $B\subset Q$ and $\ell(Q)=2^{-k}\le 12r_B$. Hence, for every $y\in Q$,

$$\|y-x_B\|\le \sqrt{N}\ell(Q)\le 12\sqrt{N}r_B.$$

This implies that $Q\subset B(x_B,12\sqrt{N}r_B)$. Enumerating $\{\mathbb{D}^{(\vartheta)}:\vartheta\in\{0,\delta\}^N\}$ as $\mathbb{D}^1,\dots,\mathbb{D}^{2^N}$, we complete the proof of Lemma 2.6. □

2.3. Nearly weakly orthonormal sequences. Rochberg and Semmes [12] introduced the notion of nearly weakly orthonormal (NWO) sequences of functions. This notion has since proved highly effective in characterizing Schatten class membership of singular integral commutators in a variety of contexts (see [2–4, 8, 9, 11–14]). The properties of NWO sequences were originally investigated in the Euclidean setting (equipped with Lebesgue measure), but the extension to spaces of homogeneous type is straightforward once a system of dyadic cubes is available in this general framework (see [8, Section 13] for a complete proof). For our purposes, we adapt the extension to the Dunkl setting $(\mathbb{R}^N,\|\cdot\|,w)$ rather than recalling the general definition of NWO sequences and the general extension to spaces of homogeneous type.

Definition 2.7. Let $\mathscr{D}$ be a system of dyadic cubes. A sequence of functions $\{e_Q\}_{Q\in\mathscr{D}}$ is an NWO sequence if

(1) e_Q is supported in cQ for some positive constant $c\in 2\mathbb{N}+1$ independent of Q;

(2) $\|e_Q\|_{L_r(\mathbb{R}^N, w(x)dx)} \le w(Q)^{\frac{1}{r}-\frac{1}{2}}$ for some $r > 2$;

Lemma 2.8. Let $0 < p < \infty$, $0 < q \le \infty$, and let $\mathscr{D}$ be a system of dyadic cubes over $\mathbb{R}^N$. Assume that

$$A = \sum_{Q\in\mathscr{D}} \lambda_Q \langle \cdot, e_Q \rangle_{L^2(\mathbb{R}^N, dw)} f_Q,$$

where $\{e_Q\}_{Q\in\mathscr{D}}$ and $\{f_Q\}_{Q\in\mathscr{D}}$ are NWO sequences and $\{\lambda_Q\}_{Q\in\mathscr{D}}$ is a sequence of scalars. We have

$$\|A\|_{S^{p,q}} \le C_{N,p,q,w} \|\{\lambda_Q\}_Q\|_{\ell^{p,q}}. \tag{2.5}$$

Proof. This property can be found in [12, (1.9)] (see also [8, Section 7]). □

3. Analytic tools for Weyl chamber geometry

Convention. Throughout this paper, we will use the following notation:

- Let $R \subset \mathbb{R}^N$ be a finite normalized reduced root system;
- Let $W := \mathrm{span}(R)$ with $N_0 := \dim W$;
- Let $\Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}$;
- Let Γ_+ be a fixed open Weyl chamber (see Definition 3.1) and Γ be its closure.

Definition 3.1. [6, Definition 2.7] A connected component of

$$\Big\{x \in \mathbb{R}^N : \langle x, v\rangle \neq 0 \quad for\ all\ v \in R\Big\} \tag{3.1}$$

is called an open Weyl chamber. Its closure is called a Weyl chamber. Define the corresponding sets of positive and negative roots by

$$R_+ := \{\alpha \in R : \ \langle x, \alpha\rangle > 0 \text{ for all } x \in \Gamma_+\}, \qquad R_- := -R_+.$$

Definition 3.2. [6, Definition 2.9] A root $\alpha \in R_+$ is called simple in R_+ if it cannot be written as

$$\alpha = \lambda_1\alpha_1 + \lambda_2\alpha_2 \quad \text{with } \lambda_1, \lambda_2 \ge 1 \text{ and } \alpha_1, \alpha_2 \in R_+.$$

The set $\Delta \subset R_+$ of all simple roots is called the simple system.

Lemma 3.3. With the notation given in Definition 3.2, Δ is a basis of W. If we denote by $\Delta = \{\alpha_1, \alpha_2, \ldots, \alpha_{N_0}\}$, then the open and closed chambers can be written as

$$\Gamma_+ = \Big\{x \in \mathbb{R}^N : \ \langle x, \alpha_i\rangle > 0, \ i = 1, \ldots, N_0\Big\}, \tag{3.2}$$

$$\Gamma = \Big\{x \in \mathbb{R}^N : \ \langle x, \alpha_i\rangle \ge 0, \ i = 1, \ldots, N_0\Big\}, \tag{3.3}$$

and every root $\alpha \in R$ admits a unique expansion

$$\alpha = \sum_{i=1}^{N_0} c_i\alpha_i$$

such that either $c_i \ge 0$ for all i (when $\alpha \in R_+$), or $c_i \le 0$ for all i (when $\alpha \in R_-$). In particular, the simple system Δ is uniquely determined by Γ_+ (equivalently, by Γ).

Proof. Let $\Delta = \{\alpha_1, \dots, \alpha_n\} \subset R_+$ be the set of simple roots in R_+. By [6, Proposition 2.10], every $\alpha \in R_+$ is a nonnegative linear combination of $\alpha_1, \dots, \alpha_n$. Since $R = R_+ \cup R_-$ (this follows from the fact that for each $\alpha \in R$, the continuous function $x \mapsto \langle x, \alpha\rangle$ does not vanish on the connected set Γ_+) and $R_- = -R_+$, it follows that $R \subset \mathrm{span}(\Delta)$ and hence,

$$W = \mathrm{span}(R) \subset \mathrm{span}(\Delta).$$

The reverse inclusion $\mathrm{span}(\Delta) \subset W$ is immediate from $\Delta \subset R \subset W$, so $\mathrm{span}(\Delta) = W$. On the other hand, it follows from [6, Theorem 2.16] that the simple roots are linearly independent. Consequently, Δ is a basis of W, and therefore $n = \dim W = N_0$. Moreover, every $\alpha \in R$ has a unique expansion $\alpha = \sum_{i=1}^{N_0} c_i\alpha_i$. If $\alpha \in R_+$, the coefficients can be chosen nonnegative by [6, Proposition 2.10]. If $\alpha \in R_- = -R_+$, then $-\alpha \in R_+$ has an expansion with all coefficients nonnegative, and multiplying by -1 gives an expansion of α with all coefficients nonpositive. The descriptions (3.2) and (3.3) follow from [6, Corollary 2.11].

To see that Δ is uniquely determined by Γ_+, note that Γ_+ determines R_+ by Definition 3.1, and the set of simple roots in R_+ is characterized intrinsically by Definition 3.2. Finally, since Γ_+ is the (unique) interior component of Γ, specifying Γ determines Γ_+ and vice versa. This completes the proof of Lemma 3.3. □

Notation 3.4. With the notation given in Lemma 3.3, denote by

$$\Delta = \{\alpha_1, \dots, \alpha_{N_0}\}$$

the simple system associated with Γ.

Lemma 3.5. *Each Weyl chamber Γ admits the decomposition*

$$\Gamma = K(\Gamma) \times W^\perp, \qquad K(\Gamma) := \{w \in W :\ \langle w, \alpha\rangle \geq 0 \text{ for all } \alpha \in \Delta\}.$$

Proof. Write $x = w + z$ with $w \in W$ and $z \in W^\perp$. Since $\alpha_i \in W$, we have $\langle x, \alpha_i\rangle = \langle w, \alpha_i\rangle$ for all $i = 1, \dots, N_0$. This, together with (3.3), implies that $x \in \Gamma$ if and only if $w \in K(\Gamma)$. This implies that $\Gamma = K(\Gamma) \times W^\perp$, and thus completes the proof. □

Corollary 3.6. *Let $\Phi_{W^\perp}$ be any linear isomorphism from $W^\perp$ to $\mathbb{R}^{N-N_0}$, and let*

$$\Phi_W : W \longrightarrow \mathbb{R}^{N_0}, \qquad \Phi_W(w) := (\langle w, \alpha_1\rangle, \dots, \langle w, \alpha_{N_0}\rangle),$$

$$\Phi : \mathbb{R}^N = W \oplus W^\perp \longrightarrow \mathbb{R}^{N_0} \oplus \mathbb{R}^{N-N_0} \cong \mathbb{R}^N, \qquad \Phi(w+z) := (\Phi_W(w),\ \Phi_{W^\perp}(z)).$$

Then Φ is a linear isomorphism and

$$\Phi(\Gamma) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

Proof. By Lemma 3.5 we have $\Gamma = K(\Gamma) \times W^\perp$, where

$$K(\Gamma) = \{w \in W :\ \langle w, \alpha_i\rangle \geq 0,\ i = 1, \dots, N_0\}.$$

Since Δ is a basis of W, the Gram matrix $G = (\langle \alpha_i, \alpha_j\rangle)_{i,j=1}^{N_0}$ is nonsingular. Hence, $\Phi_W : W \longrightarrow \mathbb{R}^{N_0}$ is a linear isomorphism. Moreover, for $w \in W$ we have $w \in K(\Gamma)$ if and only if $\Phi_W(w) \in \mathbb{R}_+^{N_0}$, so

$$\Phi_W(K(\Gamma)) = \mathbb{R}_+^{N_0}.$$

By construction, Φ is a linear isomorphism (as a direct sum of isomorphisms), and therefore

$$\Phi(\Gamma) = \Phi(K(\Gamma) \times W^{\perp}) = \Phi_W(K(\Gamma)) \times \Phi_{W^{\perp}}(W^{\perp}) = \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0},$$

which proves the desired conclusion. □

Corollary 3.7. Let Γ be an arbitrary Weyl chamber. There exists a system of dyadic cubes over Γ.

Proof. Let $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ be the linear isomorphism from Corollary 3.6 so that

$$\Phi(\Gamma) = \Omega := \mathbb{R}_+^{N_0} \times \mathbb{R}^{N-N_0}.$$

For each $k \in \mathbb{Z}$, define $\mathbb{D}^0_{\Omega,k}$ to be the family of half-open cubes

$$Q = \prod_{i=1}^{N} I_i, \qquad I_i = [m_i 2^{-k}, (m_i + 1)2^{-k}),$$

where $m_i \in \mathbb{Z}_+$ for $1 \le i \le N_0$ and $m_i \in \mathbb{Z}$ for $N_0 < i \le N$. Set

$$\mathbb{D}^0_{\Omega} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}^0_{\Omega,k}.$$

It is standard that $\mathbb{D}^0_{\Omega}$ is a system of dyadic cubes over Ω.

Next, for each $k \in \mathbb{Z}$, define

$$\mathbb{D}_{\Gamma,k} := \{\Phi^{-1}(Q) : Q \in \mathbb{D}^0_{\Omega,k}\}, \qquad \mathbb{D}_{\Gamma} := \bigcup_{k \in \mathbb{Z}} \mathbb{D}_{\Gamma,k}.$$

Since $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ is a linear isomorphism, we see that $\mathbb{D}_{\Gamma}$ is a system of dyadic cubes over Γ. □

Notation 3.8. Let Φ be the linear isomorphism given in Corollary 3.6. For every Borel set $E \subset \Omega$, define

$$\mu(E) := w(\Phi^{-1}(E)). \tag{3.4}$$

Lemma 3.9. We have

$$d\mu(u,z) = \phi(u)\, du\, dz, \quad \phi(u) = |\det(\Phi)|^{-1} \prod_{\nu \in R_+} \Big(\sum_{i=1}^{N_0} a_{\nu,i}\, u_i \Big)^{2k(\nu)}. \tag{3.5}$$

where for each $\nu \in R_+$, the coefficients $a_{\nu,i} \ge 0$ are uniquely determined by the simple-root expansion $\nu = \sum_{i=1}^{N_0} a_{\nu,i}\alpha_i$ and satisfy $(a_{\nu,1}, \dots, a_{\nu,N_0}) \ne \mathbf{0}$.

Proof. Since $\Phi(\Gamma) = \Omega$, we have $\Phi^{-1}(\Omega) = \Gamma$. Hence, for every nonnegative Borel function f on Ω,

$$\int_{\Omega} f(u,z)\, d\mu(u,z) = \int_{\Gamma} f(\Phi(x))\, dw(x).$$

We make the linear change of variables $(u,z) = \Phi(x)$, so that $x = \Phi^{-1}(u,z)$. Since Φ is linear and invertible, the Jacobian is constant and

$$dx = |\det(\Phi)|^{-1}\, du\, dz.$$

It follows from the definition of Φ that $u_i = \langle x, \alpha_i\rangle$ for $i = 1, \dots, N_0$. This implies that

$$\langle x, v\rangle = \Big\langle x, \sum_{i=1}^{N_0} a_{v,i}\alpha_i \Big\rangle = \sum_{i=1}^{N_0} a_{v,i}\langle x, \alpha_i\rangle = \sum_{i=1}^{N_0} a_{v,i}u_i. \tag{3.6}$$

Substituting these identities into the previous integral, we obtain

$$\begin{aligned}\int_\Omega f(u,z)\,d\mu(u,z) &= \int_\Gamma f(\Phi(x))\Big(\prod_{v\in R_+} |\langle x, v\rangle|^{2k(v)}\Big)\,dx \\ &= \int_\Omega f(u,z)\,|\det(\Phi)|^{-1} \prod_{v\in R_+}\Big(\sum_{i=1}^{N_0} a_{v,i}u_i\Big)^{2k(v)}\,du\,dz.\end{aligned}$$

Therefore, (3.5) holds. This completes the proof of Lemma 3.9. □

Lemma 3.10. Let Γ_1 and Γ_2 be two Weyl chambers. Then there exist an integer $M \geq 0$ and $\alpha_1, \dots, \alpha_M \in R$ such that, if we set $\Gamma^{(0)} := \Gamma_2$ and

$$\Gamma^{(k)} := \sigma_{\alpha_k}(\Gamma^{(k-1)}), \qquad k = 1, \dots, M,$$

then the following hold:

(1) $\alpha_k \in \Delta(\Gamma^{(k-1)})$ for each $k = 1, \dots, M$;
(2) $\Gamma^{(M)} = \Gamma_1$.

Proof. If $\Gamma_1 = \Gamma_2$, we take $M = 0$ and there is nothing to prove. Assume $\Gamma_1 \neq \Gamma_2$. Since the Weyl group G acts transitively on the set of Weyl chambers (see [6, Theorem 2.12 (2)]), there exists $g \in G$ such that $\Gamma_1 = g(\Gamma_2)$.

Let $\Delta(\Gamma_2) = \{\beta_1, \beta_2, \dots, \beta_{N_0}\}$ be the simple system associated with Γ_2. By [6, Theorem 2.12(4)], the Weyl group G is generated by the simple reflections $\sigma_{\beta_1}, \sigma_{\beta_2}, \dots, \sigma_{\beta_{N_0}}$. Hence, there exist indices $i_1, \dots, i_M \in \{1, \dots, N_0\}$ such that

$$g = \sigma_{\beta_{i_1}}\sigma_{\beta_{i_2}}\cdots\sigma_{\beta_{i_M}}.$$

Define

$$w_0 := \mathrm{Id}, \qquad w_k := \sigma_{\beta_{i_1}}\cdots\sigma_{\beta_{i_k}} \quad (k = 1, \dots, M),$$

and set

$$\Gamma^{(k)} := w_k(\Gamma_2) \qquad (k = 0, 1, \dots, M).$$

It follows that $\Gamma^{(0)} = \Gamma_2$ and $\Gamma^{(M)} = w_M(\Gamma_2) = g(\Gamma_2) = \Gamma_1$, which implies the statement (2).

For each $k = 1, \dots, M$, define

$$\alpha_k := w_{k-1}(\beta_{i_k}) \in R.$$

The rest of the proof is divided into two steps.

- Step 1. Verify that $\Gamma^{(k)} = \sigma_{\alpha_k}(\Gamma^{(k-1)})$.

Since each $w \in G$ is orthogonal, for any root $\beta \in R$ one has the conjugation identity

$$w\,\sigma_\beta\,w^{-1} = \sigma_{w(\beta)}, \qquad w \in G.$$

Applying this with $w = w_{k-1}$ and $\beta = \beta_{i_k}$ yields

$$\sigma_{\alpha_k} = \sigma_{w_{k-1}(\beta_{i_k})} = w_{k-1}\sigma_{\beta_{i_k}}w_{k-1}^{-1}.$$

Therefore,

$$\sigma_{\alpha_k}(\Gamma^{(k-1)}) = w_{k-1}\sigma_{\beta_{i_k}} w_{k-1}^{-1}(w_{k-1}(\Gamma_2)) = w_{k-1}\sigma_{\beta_{i_k}}(\Gamma_2) = w_k(\Gamma_2) = \Gamma^{(k)}.$$

- Step 2. Verify that $\alpha_k \in \Delta(\Gamma^{(k-1)})$.

Recall that if Γ is a Weyl chamber associated with simple system $\Delta(\Gamma)$, then

$$\Gamma = \{x \in \mathbb{R}^N : \ \langle x, \gamma \rangle \geq 0 \text{ for all } \gamma \in \Delta(\Gamma)\}.$$

Applying $w_{k-1} \in G$ to the above description for Γ_2 and using $\langle w_{k-1}x, \beta \rangle = \langle x, w_{k-1}^{-1}\beta \rangle$, we obtain

$$\Gamma^{(k-1)} = w_{k-1}(\Gamma_2) = \Big\{x \in \mathbb{R}^N : \ \langle x, w_{k-1}(\beta_j) \rangle \geq 0 \text{ for } j = 1, \dots, N_0\Big\}.$$

By the uniqueness of the simple system associated with a Weyl chamber (see Lemma 3.3), we conclude that

$$\Delta(\Gamma^{(k-1)}) = \Delta\big(w_{k-1}(\Gamma_2)\big) = w_{k-1}(\Delta(\Gamma_2)) = \{w_{k-1}(\beta_1), \dots, w_{k-1}(\beta_{N_0})\}.$$

In particular, $\alpha_k = w_{k-1}(\beta_{i_k}) \in \Delta(\Gamma^{(k-1)})$, proving the first statement (1).

This completes the proof of Lemma 3.10. □

Definition 3.11. We say that $R' \subset \Omega$ is an axis–parallel rectangle if

$$R' := I_1 \times \cdots \times I_N,$$

with $I_i = [I_i^-, I_i^+] \subset [0, \infty)$ for $1 \leq i \leq N_0$ and $I_i = [I_i^-, I_i^+] \subset \mathbb{R}$ for $i > N_0$.

Denote by

$$B_\Omega(x_0, r) := \{x \in \Omega : \ \|x - x_0\| < r\}.$$

The following lemma demonstrates that $(\Omega, \|\cdot\|, d\mu)$ is a doubling metric measure space.

Lemma 3.12. There exists a constant $C \geq 1$ such that for every $x_0 = (u_0, z_0) \in \Omega$ and every $r > 0$,

$$C^{-1}\, r^N \prod_{\nu \in R_+} \Big(L_{\nu,0} + r\Big)^{2k(\nu)} \ \leq\ \mu(B_\Omega(x_0, r)) \ \leq\ C\, r^N \prod_{\nu \in R_+} \Big(L_{\nu,0} + r\Big)^{2k(\nu)}, \tag{3.7}$$

where

$$L_\nu(u) := \sum_{i=1}^{N_0} a_{\nu,i} u_i, \qquad L_{\nu,0} := L_\nu(u_0) = \sum_{i=1}^{N_0} a_{\nu,i}(u_0)_i.$$

In particular, there exists a constant $C_{\mathrm{dbl},\Omega} \geq 1$ such that for every $x \in \Omega$ and every $r > 0$,

$$\mu(B_\Omega(x, 2r)) \ \leq\ C_{\mathrm{dbl},\Omega}\, \mu(B_\Omega(x, r)).$$

Proof. Fix $x_0 = (u_0, z_0) \in \Omega$ and $r > 0$. For each $\nu \in R_+$ define the linear form

$$L_\nu(u) := a_\nu \cdot u, \qquad a_\nu := (a_{\nu,1}, \dots, a_{\nu,N_0}) \in \mathbb{R}_+^{N_0} \setminus \{0\}.$$

Since R_+ is finite and each $a_\nu \neq 0$, we have

$$L_* := \max_{\nu \in R_+} \|a_\nu\| < \infty, \qquad m_* := \min_{\nu \in R_+} \sum_{i=1}^{N_0} a_{\nu,i} > 0.$$

Upper bound. Let $x = (u, z) \in B_\Omega(x_0, r)$. Then

$$\|u - u_0\| \;\leq\; \|x - x_0\| \;\leq\; r.$$

Hence, for every $\nu \in R_+$,

$$|L_\nu(u) - L_{\nu,0}| = \left|a_\nu \cdot (u - u_0)\right| \leq \|a_\nu\| \cdot \|u - u_0\| \leq L_* \, r.$$

Since $L_\nu(u) \geq 0$ on Ω (all coefficients and coordinates are nonnegative), we obtain

$$L_\nu(u) \leq L_{\nu,0} + L_* r \leq C_1\,(L_{\nu,0} + r), \qquad C_1 := \max\{1, L_*\}.$$

Therefore,

$$\phi(u) = \prod_{\nu \in R_+} L_\nu(u)^{2k(\nu)} \leq C_1^{2\sum_{\nu \in R_+} k(\nu)} \prod_{\nu \in R_+} (L_{\nu,0} + r)^{2k(\nu)} =: C_2 \prod_{\nu \in R_+} (L_{\nu,0} + r)^{2k(\nu)}.$$

Integrating over $B_\Omega(x_0, r)$ and using $|B_\Omega(x_0, r)| \leq |B(x_0, r)| = \omega_N r^N$, where ω_N is the volume of the unit ball in $\mathbb{R}^N$, gives

$$\mu(B_\Omega(x_0, r)) = \int_{B_\Omega(x_0,r)} \phi(u)\,dx \leq C_2\,|B(x_0, r)| \prod_{\nu \in R_+} (L_{\nu,0} + r)^{2k(\nu)} \leq C_3\, r^N \prod_{\nu \in R_+} (L_{\nu,0} + r)^{2k(\nu)},$$

with $C_3 := C_2\,\omega_N$.

Lower bound. We construct a smaller Euclidean ball B_c contained in $B_\Omega(x_0, r)$ on which ϕ admits a uniform positive lower bound in terms of $\prod_{\nu \in R_+} (L_{\nu,0} + r)^{2k(\nu)}$. To this end, we first choose

$$\lambda := 1 + \frac{L_*}{m_*} > 1, \qquad \eta := \frac{1}{4(\sqrt{N_0}\,\lambda + 1)}, \qquad \theta := \lambda\,\eta.$$

These parameters depend only on L_* and m_*, and satisfy

$$\theta > \eta > 0, \qquad \sqrt{N_0}\,\theta + \eta = \eta(\sqrt{N_0}\lambda + 1) = \frac{1}{4} < 1.$$

Let $\mathbf{1}_{N_0} := (1, \ldots, 1) \in \mathbb{R}^{N_0}$ and define the center

$$x_c := (u_c, z_c) := (u_0 + \theta r\,\mathbf{1}_{N_0},\; z_0), \qquad B_c := B(x_c, \eta r).$$

We claim that $B_c \subset B_\Omega(x_0, r)$. To see this inclusion, we first apply the triangle inequality to see that for any $x \in B_c$,

$$\|x - x_0\| \leq \|x - x_c\| + \|x_c - x_0\| \leq \eta r + \theta r \sqrt{N_0} \leq r,$$

since $\sqrt{N_0}\theta + \eta < 1$. Hence $B_c \subset B(x_0, r)$. Moreover, for any $x = (u, z) \in B_c$, we have $\|u - u_c\| \leq \|x - x_c\| \leq \eta r$. This implies that for each $1 \leq i \leq N_0$,

$$u_i \geq (u_c)_i - \eta r = (u_0)_i + (\theta - \eta) r \geq 0,$$

since $u_0 \in \mathbb{R}_+^{N_0}$ and $\theta \geq \eta$. Thus $u \in \mathbb{R}_+^{N_0}$ and $x \in \Omega$. Therefore, $B_c \subset B_\Omega(x_0, r)$. This finishes the proof of the claim.

Next we obtain a uniform lower bound for $L_\nu(u)$ on B_c. Write any $x = (u, z) \in B_c$ as

$$u = u_0 + \theta r\,\mathbf{1}_{N_0} + \Delta_u, \qquad z = z_0 + \Delta_z, \qquad \|\Delta_u\|^2 + \|\Delta_z\|^2 \leq (\eta r)^2,$$

so in particular $\|\Delta_u\| \le \eta r$. Then for all $x = (u, z) \in B_c$ and all $\nu \in R_+$,

$$\begin{aligned}L_\nu(u) &= a_\nu \cdot u = a_\nu \cdot u_0 + \theta r\, a_\nu \cdot \mathbf{1}_{N_0} + a_\nu \cdot \Delta_u \\ &\ge L_{\nu,0} + \theta r \sum_{i=1}^{N_0} a_{\nu,i} - \|a_\nu\|\,\|\Delta_u\| \ge L_{\nu,0} + rK_0 \ge c_0(L_{\nu,0} + r).\end{aligned}$$

with

$$K_0 := \theta m_* - \eta L_* = \eta(\lambda m_* - L_*) = \eta m_* > 0, \qquad c_0 := \min\{1, K_0\} \in (0, 1].$$

Consequently, for all $u \in B_c$,

$$\phi(u) = \prod_{\nu\in R_+} L_\nu(u)^{2k(\nu)} \ge c_0^{2\sum_{\nu\in R_+} k(\nu)} \prod_{\nu\in R_+} (L_{\nu,0} + r)^{2k(\nu)} =: C_4 \prod_{\nu\in R_+} (L_{\nu,0} + r)^{2k(\nu)}.$$

This, in combination with the fact that $B_c \subset B_\Omega(x_0, r)$, yields

$$\mu(B_\Omega(x_0, r)) \ge \mu(B_c) = \int_{B_c} \phi(u)\, dx \ge C_4\, |B_c| \prod_{\nu\in R_+} (L_{\nu,0} + r)^{2k(\nu)} = C_5\, r^N \prod_{\nu\in R_+} (L_{\nu,0} + r)^{2k(\nu)},$$

where $C_5 := C_4\, \omega_N \eta^N > 0$.

The second statement follows immediately from the first one. Indeed, by the first statement,

$$\mu(B_\Omega(x_0, 2r)) \le C_3\, (2r)^N \prod_{\nu\in R_+} (L_{\nu,0}+2r)^{2k(\nu)} \le C_3\, 2^{N+\mathcal{K}}\, r^N \prod_{\nu\in R_+} (L_{\nu,0}+r)^{2k(\nu)} \le C_3 C_5^{-1}\, 2^{N+\mathcal{K}}\, \mu(B_\Omega(x_0, r)),$$

where we set $\mathcal{K} := 2\sum_{\nu\in R_+} k(\nu)$. Therefore, $(\Omega, \|\cdot\|, \mu)$ is doubling with the doubling constant $C_{\mathrm{dbl},\Omega} := C_3 C_5^{-1}\, 2^{N+\mathcal{K}}$. This finishes the proof of Lemma 3.12. □

Let $\Phi : \mathbb{R}^N \longrightarrow \mathbb{R}^N$ be the linear isomorphism given in Corollary 3.6 such that $\Phi(\Gamma) = \Omega$. Since Φ is linear and invertible, there exist positive constants c_Φ and C_Φ such that for any $x, y \in \mathbb{R}^N$,

$$c_\Phi\, \|x - y\| \ \le\ \|\Phi(x) - \Phi(y)\| \ \le\ C_\Phi\, \|x - y\|. \tag{3.8}$$

Denote by $B_\Gamma(x, r) := \{y \in \Gamma :\ \|y - x\| < r\}$. It follows from (3.8) that for any $x_0 \in \Gamma$ and $r > 0$,

$$B_\Omega(\Phi(x_0), c_\Phi r) \subset \Phi(B_\Gamma(x_0, r)) \subset B_\Omega(\Phi(x_0), C_\Phi r). \tag{3.9}$$

This, together with Lemma 3.12, yields

$$\mu(\Phi(B_\Gamma(x_0, r))) \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)). \tag{3.10}$$

Lemma 3.13. For all $x_0 \in \Gamma$ and $r > 0$, one can find an axis–parallel rectangle R' on Ω such that

$$B_\Omega(\Phi(x_0), C_\Phi r) \subset R' \subset B_\Omega(\Phi(x_0),\ \sqrt{N} C_\Phi r). \tag{3.11}$$

Furthermore, we have

$$\mu(R') \simeq \mu(B_\Omega(\Phi(x_0), C_\Phi r)). \tag{3.12}$$

Proof. We write $(u_0, z_0) := \Phi(x_0) \in \Omega$. and define the axis–parallel rectangle

$$R' := \Big(\prod_{i=1}^{N_0} [\max\{0, (u_0)_i - C_\Phi r\},\ (u_0)_i + C_\Phi r] \Big) \times \Big(\prod_{j=1}^{N-N_0} [(z_0)_j - C_\Phi r,\ (z_0)_j + C_\Phi r] \Big). \tag{3.13}$$

On the one hand, it follows immediately that $B_\Omega(\Phi(x_0), C_\Phi r) \subset R'$. On the other hand, since R' is a rectangle with half-side lengths bounded by $C_\Phi r$ in each coordinate, we have

$$R' \subset B_\Omega(\Phi(x_0),\ \sqrt{N}\, C_\Phi r).$$

The second statement follows from the first one together with Lemma 3.12. The proof of Lemma 3.13 is finished. □

The following theorem demonstrates that $(\Gamma, \|\cdot\|, dw)$ is a doubling metric measure space.

Theorem 3.14. There exists a constant $C \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$C^{-1}\, r^N \prod_{v \in R_+} (|\langle x, v\rangle| + r)^{2k(v)} \ \leq\ w(B_\Gamma(x, r)) \ \leq\ C\, r^N \prod_{v \in R_+} (|\langle x, v\rangle| + r)^{2k(v)}. \tag{3.14}$$

In particular, there exists a constant $C_{\mathrm{dbl},\Gamma} \geq 1$ such that for every $x \in \Gamma$ and every $r > 0$,

$$w(B_\Gamma(x, 2r)) \ \leq\ C_{\mathrm{dbl},\Gamma}\, w(B_\Gamma(x, r)).$$

Moreover, for all $x \in \Gamma$ and $r > 0$, we have

$$w(B_\Gamma(x, r)) \sim w(B(x, r)). \tag{3.15}$$

Proof. The main strategy of the proof is to transfer the ball estimates from $(\Omega, \|\cdot\|, \mu)$ back to $(\Gamma, \|\cdot\|, dw)$. Fix $x \in \Gamma$ and $r > 0$. It follows from the equality (3.4) that

$$w(B_\Gamma(x, r)) = \mu(\Phi(B_\Gamma(x, r))). \tag{3.16}$$

Applying μ to the inclusions (3.9) and using (3.16) yields

$$\mu(B_\Omega(\Phi(x), c_\Phi r)) \ \leq\ w(B_\Gamma(x, r)) \ \leq\ \mu(B_\Omega(\Phi(x), C_\Phi r)). \tag{3.17}$$

Write $\Phi(x) = (u, z) \in \Omega$. For each $v \in R_+$ recall the linear form $L_{v,0} := \sum_{i=1}^{N_0} a_{v,i} u_i$. Combining the inequality (3.17) with Lemma 3.12 and the equality (3.6), we obtain that

$$w(B_\Gamma(x, r)) \ \simeq\ r^N \prod_{v \in R_+} (L_{v,0} + r)^{2k(v)} = r^N \prod_{v \in R_+} (|\langle x, v\rangle| + r)^{2k(v)}. \tag{3.18}$$

This proves (3.14). The second statement follows immediately from (3.14). The third statement follows from the inequalities (3.14) and (2.1). This completes the proof of Theorem 3.14. □

4. Cwikel's estimate in $N > 2$

Lemma 4.1. Assume that $N > 1$. Then for every pair of multi-indices α, β, there exists a constant $C_{\alpha,\beta} > 0$ such that

$$\left|\partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}(x, y)\right| \leq C_{\alpha,\beta} \frac{d(x, y)^{1-|\alpha|-|\beta|}}{V(x, y, d(x, y))} \tag{4.19}$$

for all $x, y \in \mathbb{R}^N$ satisfying $d(x, y) > 0$.

Proof. Let $h_t(x,y)$ denote the integral kernel of the Dunkl heat semigroup $e^{t\Delta_{\mathrm{Dunkl}}}$. By [1, Theorem 4.1(d)], for every pair of multi-indices α,β, there exist constants $C_{\alpha,\beta},c_{\alpha,\beta}>0$ such that

$$\left|\partial_x^\alpha\partial_y^\beta h_t(x,y)\right|\le C_{\alpha,\beta}t^{-\frac{|\alpha|+|\beta|}{2}}V(x,y,\sqrt{t})^{-1}\exp\left(-c_{\alpha,\beta}\frac{d(x,y)^2}{t}\right)\tag{4.20}$$

for all $t>0$ and $x,y\in\mathbb{R}^N$.

By the spectral functional calculus,

$$(-\Delta_{\mathrm{Dunkl}})^{-1/2}=\frac{1}{\sqrt{\pi}}\int_0^\infty e^{t\Delta_{\mathrm{Dunkl}}}\,\frac{dt}{\sqrt{t}}.$$

Consequently, for $d(x,y)>0$,

$$K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}(x,y)=\frac{1}{\sqrt{\pi}}\int_0^\infty h_t(x,y)\,\frac{dt}{\sqrt{t}}.\tag{4.21}$$

It follows that $K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}$ is smooth whenever $d(x,y)>0$ and

$$\partial_x^\alpha\partial_y^\beta K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}(x,y)=\frac{1}{\sqrt{\pi}}\int_0^\infty \partial_x^\alpha\partial_y^\beta h_t(x,y)\,\frac{dt}{\sqrt{t}}.\tag{4.22}$$

Set $m:=|\alpha|+|\beta|$. Fix $x,y\in\mathbb{R}^N$ with $d(x,y)>0$. By (4.20) and (4.22),

$$\left|\partial_x^\alpha\partial_y^\beta K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}(x,y)\right|\lesssim\int_0^\infty t^{-\frac{m+1}{2}}V(x,y,\sqrt{t})^{-1}\exp\left(-c\frac{d(x,y)^2}{t}\right)dt.\tag{4.23}$$

Suppose first that $0<t\le d(x,y)^2$. Then $\sqrt{t}\le d(x,y)$. By the upper estimate in (2.2), for $z=x,y$,

$$\frac{w(B(z,d(x,y)))}{w(B(z,\sqrt{t}))}\lesssim\left(\frac{d(x,y)}{\sqrt{t}}\right)^{\mathbf{N}}.$$

Equivalently,

$$w(B(z,\sqrt{t}))\gtrsim\left(\frac{\sqrt{t}}{d(x,y)}\right)^{\mathbf{N}}w(B(z,d(x,y))).$$

Taking the maximum over $z=x,y$ gives

$$V(x,y,\sqrt{t})\gtrsim\left(\frac{\sqrt{t}}{d(x,y)}\right)^{\mathbf{N}}V(x,y,d(x,y)).\tag{4.24}$$

Therefore,

$$\begin{aligned}&\int_0^{d(x,y)^2}t^{-\frac{m+1}{2}}V(x,y,\sqrt{t})^{-1}\exp\left(-c\frac{d(x,y)^2}{t}\right)dt\\&\qquad\lesssim\frac{d(x,y)^{\mathbf{N}}}{V(x,y,d(x,y))}\int_0^{d(x,y)^2}t^{-\frac{\mathbf{N}+m+1}{2}}\exp\left(-c\frac{d(x,y)^2}{t}\right)dt.\end{aligned}$$

Using again the change of variables $s=\frac{d(x,y)^2}{t}$, we obtain

$$\int_0^{d(x,y)^2}t^{-\frac{\mathbf{N}+m+1}{2}}\exp\left(-c\frac{d(x,y)^2}{t}\right)dt=d(x,y)^{1-\mathbf{N}-m}\int_1^\infty s^{\frac{\mathbf{N}+m-3}{2}}e^{-cs}\,ds\lesssim d(x,y)^{1-\mathbf{N}-m}.$$

Consequently,

$$\int_0^{d(x,y)^2} t^{-\frac{m+1}{2}} V(x,y,\sqrt{t})^{-1} \exp\left(-c\frac{d(x,y)^2}{t}\right) dt \lesssim \frac{d(x,y)^{1-m}}{V(x,y,d(x,y))}. \tag{4.25}$$

Suppose next that $t \geq d(x,y)^2$. Then $\sqrt{t} \geq d(x,y)$. By the lower estimate in (2.2), for $z = x, y$,

$$\frac{w(B(z,\sqrt{t}))}{w(B(z,d(x,y)))} \gtrsim \left(\frac{\sqrt{t}}{d(x,y)}\right)^N.$$

It follows that

$$V(x,y,\sqrt{t}) \gtrsim \left(\frac{\sqrt{t}}{d(x,y)}\right)^N V(x,y,d(x,y)). \tag{4.26}$$

Hence

$$\int_{d(x,y)^2}^{\infty} t^{-\frac{m+1}{2}} V(x,y,\sqrt{t})^{-1} \exp\left(-c\frac{d(x,y)^2}{t}\right) dt \lesssim \frac{d(x,y)^N}{V(x,y,d(x,y))} \int_{d(x,y)^2}^{\infty} t^{-\frac{N+m+1}{2}}\, dt.$$

Since $N > 1$ and $m \geq 0$, we have $N + m > 1$, and therefore

$$\int_{d(x,y)^2}^{\infty} t^{-\frac{N+m+1}{2}}\, dt = \frac{2}{N+m-1} d(x,y)^{1-N-m}.$$

Thus

$$\int_{d(x,y)^2}^{\infty} t^{-\frac{m+1}{2}} V(x,y,\sqrt{t})^{-1} \exp\left(-c\frac{d(x,y)^2}{t}\right) dt \lesssim \frac{d(x,y)^{1-m}}{V(x,y,d(x,y))}. \tag{4.27}$$

Combining (4.23), (4.25), and (4.27), we obtain

$$\left|\partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}(x,y)\right| \lesssim \frac{d(x,y)^{1-m}}{V(x,y,d(x,y))}.$$

Since $m = |\alpha| + |\beta|$, this gives

$$\left|\partial_x^\alpha \partial_y^\beta K_{(-\Delta_{\mathrm{Dunkl}})^{-1/2}}(x,y)\right| \lesssim \frac{d(x,y)^{1-|\alpha|-|\beta|}}{V(x,y,d(x,y))},$$

which is exactly (4.19). □

Lemma 4.2. Assume that $N > 2$. There exists a constant $C_{N,R,k} > 0$ such that, for every $f \in L_N(\mathbb{R}^N, dx)$,

$$M_f(-\Delta_{\mathrm{Dunkl}})^{-\frac{1}{2}} \in \mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))$$

and

$$\left\|M_f(-\Delta_{\mathrm{Dunkl}})^{-\frac{1}{2}}\right\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \leq C_{N,R,k}\|f\|_{L_N(\mathbb{R}^N,dx)}. \tag{4.28}$$

Proof. For $g \in G$, let $U_g : \xi \to \xi \circ g^{-1}$. For a chamber Γ and $g \in G$, define an integral operator

$$V_{\Gamma,g} = M_{\chi_\Gamma}(-\Delta_{\mathrm{Dunkl}})^{-\frac{1}{2}} M_{\chi_{g\Gamma}} U_g.$$

By Lemma 4.1, each operator $V_{\Gamma,g}$ satisfies the assumptions in Theorem 5.1. We, obviously, have

$$M_f(-\Delta_{\mathrm{Dunkl}})^{-\frac{1}{2}} = \sum_{\Gamma}\sum_{g\in G} M_f V_{\Gamma,g} \cdot U_{g^{-1}}.$$

The assertion follows now from Theorem 5.1. □

5. Better Cwikel estimate

Theorem 5.1. Let Γ be a chamber. Let V be an integral operator on $L_2(\Gamma, w(x)dx)$ with integral kernel K satisfying the conditions

$$|\partial_x^\alpha \partial_y^\beta K(x,y)| \leq C_{\alpha,\beta} \frac{|x-y|^{1-|\alpha|_1-|\beta|_1}}{V(x,|x-y|)}.$$

We have

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(\Gamma,w(x)dx)} \leq c_K \|f\|_{L_N(\Gamma)}.$$

Notation 5.2. Fix a closed Weyl chamber Γ and let $\Phi : \mathbb{R}^N \to \mathbb{R}^N$ be the linear isomorphism appearing in Corollary 3.6.

Let $\mathbb{D}_\Gamma$ be the dyadic system over Γ constructed in Corollary 3.7. If $I \in \mathbb{D}_\Gamma$ and if $Q = \Phi(I)$, then we denote

$$d_I = \ell(Q).$$

Lemma 5.3. There exists a smooth function ψ supported on $[-1,1]^N \backslash (-\frac{1}{4}, \frac{1}{4})^N$ such that

$$1 = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \chi_{3I}(y) \psi(\frac{\Phi(x)-\Phi(y)}{d_I}).$$

Proof. Choose $\rho \in C_c^\infty(\mathbb{R}^N)$ such that $\rho = 1$ on $(-\frac{1}{2}, \frac{1}{2})^N$ and $\rho = 0$ outside $(-1,1)^N$. Set $\psi(z) = \rho(z) - \rho(2z)$. Clearly,

$$\operatorname{supp}(\psi) \subset \left\{ z \in \mathbb{R}^N : \quad \frac{1}{4} \leq |z|_\infty \leq 1 \right\}. \tag{5.29}$$

For every $z \neq 0$, we have

$$\sum_{j \in \mathbb{Z}} \psi(2^j z) = 1. \tag{5.30}$$

For every $x \in \Gamma$, there exists a unique $I \in \mathbb{D}_{\Gamma,j}$ containing x. For this I, we have $d_I = 2^{-j}$. We have

$$1 = \sum_{j \in \mathbb{Z}} \psi(2^j(\Phi(x)-\Phi(y))) = \sum_{j \in \mathbb{Z}} (\sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x)) \psi(2^j(\Phi(x)-\Phi(y))) =$$

$$= \sum_{j \in \mathbb{Z}} \sum_{I \in \mathbb{D}_{\Gamma,j}} \chi_I(x) \psi(\frac{\Phi(x)-\Phi(y)}{d_I}) = \sum_{I \in \mathbb{D}_\Gamma} \chi_I(x) \psi(\frac{\Phi(x)-\Phi(y)}{d_I}).$$

If $x \in I$ and $\psi(\frac{\Phi(x)-\Phi(y)}{d_I}) \neq 0$, then

$$\Phi(x) \in \Phi(I) = Q, \quad |\Phi(x)-\Phi(y)| \leq d_I = \ell(Q).$$

We obviously have $|\Phi(x) - c_Q|_\infty \leq \frac{1}{2}\ell(Q)$ and $|\Phi(x)-\Phi(y)|_\infty \leq \ell(Q)$. Thus, $|\Phi(y) - c_Q| \leq \frac{3}{2}\ell(Q)$. In other words, $\Phi(y) \in 3Q$ or, equivalently, $y \in 3I$. Thus,

$$\chi_I(x)\psi(\frac{\Phi(x)-\Phi(y)}{d_I}) = \chi_I(x)\chi_{3I}(y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I}).$$

This completes the proof. □

Lemma 5.4. Assume that $N>1$ and $2<r<\infty$. Let $f\in L_r^{\rm loc}(\mathbb{R}^N,w(x)dx)$. There exist functions $\{e_{k,l,I}\}_{k,l\in\mathbb{Z}^N,I\in\mathbb{D}_\Gamma}$, $\{f_{k,l,I}\}_{k,l\in\mathbb{Z}^N,I\in\mathbb{D}_\Gamma}$, such that

(1) $e_{k,l,I}$ is supported in I and $f_{k,l,I}$ is supported in $3I$;

(2)
$$\|e_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)}=\|f\|_{L_r(\mathbb{R}^N,w(x)dx)},\quad \|f_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)}\le c_{K,\kappa}d_Iw(I)^{\frac1r-1};$$

(3) For almost every $x,y\in\Gamma$, we have
$$f(x)K(x,y)=\sum_{k,l\in\mathbb{Z}^N}(1+|k|^2+|l|^2)^{-2N-1}\sum_{I\in\mathbb{D}_\Gamma}e_{k,l,I}(x)f_{k,l,I}(y). \tag{5.31}$$

Proof. We have
$$K(x,y)=\sum_{I\in\mathbb{D}_\Gamma}\chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I})=$$
$$=\sum_{\substack{I\in\mathbb{D}_\Gamma\\3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I})+\sum_{\substack{I\in\mathbb{D}_\Gamma\\3I\not\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I}).$$

We will only consider the first sum. The second some can be treated similarly, but the notations become heavier. feel free to write the expression for the second sum if you think it is needed. the only difference is that the domain of the map K_I becomes more complicated.

For every $I\in\mathbb{D}_\Gamma$ with $3I\subset\Gamma$, define the map
$$K_I:[-\frac12,\frac12]^N\times[-\frac32,\frac32]^N\to\mathbb{C}$$
by setting
$$K_I(a,b)=K(c_I+d_I\Phi^{-1}(a),c_I+d_I\Phi^{-1}(b))\psi(a-b),\quad a\in[-\frac12,\frac12]^N,\quad b\in[-\frac32,\frac32]^N.$$

It follows from the assumption on K that feel free to prove this if you think this requires a proof. however, do not prove it here — make a separate lemma instead.
$$\|K_I\|_{C^{4N+2}([-\frac12,\frac12]^N\times[-\frac32,\frac32]^N)}\le c_K\frac{d_I}{w(I)}.$$
Hence, there exists an extension $K_I:\mathbb{T}^N\times\mathbb{T}^N\to\mathbb{C}$ such that
$$\|K_I\|_{C^{4N+2}(\mathbb{T}^N\times\mathbb{T}^N)}\le c_K'\frac{d_I}{w(I)}.$$
In particular,
$$\|K_I\|_{W^{4N+2,2}(\mathbb{T}^N\times\mathbb{T}^N)}\le c_K''\frac{d_I}{w(I)}.$$
We may, therefore, write
$$K_I(a,b)=\sum_{k,l\in\mathbb{Z}^N}{\rm coef}_{k,l,I}e^{i\langle k,a\rangle}e^{i\langle l,b\rangle},\quad a\in[-\frac12,\frac12]^N,\quad b\in[-\frac32,\frac32]^N,$$
where
$$\sum_{k,l\in\mathbb{Z}^N}(1+|k|^2+|l|^2)^{4N+2}|{\rm coef}_{k,l,I}|^2\le(c_K''\frac{d_I}{w(I)})^2.$$

In particular,

$$|\mathrm{coef}_{k,l,I}| \leq c''_K(1+|k|^2+|l|^2)^{-2N-1}\frac{d_I}{w(I)}.$$

This allows us to write

$$\sum_{\substack{I\in\mathbb{D}_\Gamma\\ 3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K(x,y)\psi(\frac{\Phi(x)-\Phi(y)}{d_I}) = \sum_{\substack{I\in\mathbb{D}_\Gamma\\ 3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)K_I(\frac{\Phi^{-1}(x-c_I)}{d_I},\frac{\Phi^{-1}(x)-c_I}{d_I}) =$$

$$= \sum_{\substack{I\in\mathbb{D}_\Gamma\\ 3I\subset\Gamma}}\chi_I(x)\chi_{3I}(y)\sum_{k,l\in\mathbb{Z}^N}\mathrm{coef}_{k,l,I}e^{i\langle k,\frac{\Phi^{-1}(x-c_I)}{d_I}\rangle}e^{i\langle l,\frac{\Phi^{-1}(y-c_I)}{d_I}\rangle} =$$

$$= \sum_{k,l\in\mathbb{Z}^N}(1+|k|^2+|l|^2)^{-2N-1}\sum_{\substack{I\in\mathbb{D}_\Gamma\\ 3I\subset\Gamma}}(1+|k|^2+|l|^2)^{2N+1}\mathrm{coef}_{k,l,I}\cdot\chi_I(x)e^{i\langle k,\frac{\Phi^{-1}(x-c_I)}{d_I}\rangle}\cdot\chi_{3I}(y)e^{i\langle l,\frac{\Phi^{-1}(y-c_I)}{d_I}\rangle}.$$

We now set

$$e_{k,l,I}(x) = f(x)\chi_I(x)e^{i\langle k,\frac{\Phi^{-1}(x-c_I)}{d_I}\rangle},$$

$$f_{k,l,I}(y) = (1+|k|^2+|l|^2)^{2N+1}\mathrm{coef}_{k,l,I}\cdot\chi_{3I}(y)e^{i\langle l,\frac{\Phi^{-1}(y-c_I)}{d_I}\rangle}.$$

Obviously,

$$\mathrm{supp}(e_{k,l,I})\subset I, \quad \mathrm{subset}(f_{k,l,I})\subset 3I,$$

$$\|e_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)} = \|f\|_{L_r(I,w(x)dx)},$$

$$\|f_{k,l,I}\|_{L_r(\mathbb{R}^N,w(x)dx)} = (1+|k|^2+|l|^2)^{2N+1}|\mathrm{coef}_{I,k,l}|\cdot w(3I)^{\frac{1}{r}} \leq c''_K\frac{d_I}{w(I)}\cdot w(3I)^{\frac{1}{r}} \leq c_{K,\kappa}d_Iw(I)^{\frac{1}{r}-1}.$$

□

Lemma 5.5. We have

$$\|M_fV\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N,w(x)dx))} \leq c_{K,\kappa}\Big\|\Big\{d_Iw(I)^{-\frac{1}{r}}\|f\|_{L_r(I,w(x)dx)}\Big\}_{I\in\mathbb{D}_\Gamma}\Big\|_{l_{N,\infty}}.$$

Proof. The assertion follows from Lemma 5.4, Lemma 2.8 and the triangle inequality. feel free to expand the argument if needed. □

Lemma 5.6. Let $1<s<\infty$, and let $\mathbb{D}^0$ be the standard dyadic grid in $\mathbb{R}^N$. Suppose that $F\in L^s(\mathbb{R}^N)$, For every $Q\in\mathbb{D}^0$, set

$$b_Q := |Q|^{\frac{1}{s}-1}\int_Q|F(x)|\,dx = |Q|^{-1/s'}\int_Q|F(x)|\,dx,$$

Then

$$\#\{Q\in\mathbb{D}^0 : b_Q>t\} \lesssim_{N,s} t^{-s}\|F\|^s_{L^s(\mathbb{R}^N)}, \qquad t>0. \tag{5.32}$$

Equivalently,

$$\Big\|\{b_Q\}_{Q\in\mathbb{D}^0}\Big\|_{\ell^{s,\infty}} \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)}.$$

Proof. Let $\mathcal{E} \subset \mathbb{D}^0$ be an arbitrary finite collection of dyadic cubes, and define

$$G_{\mathcal{E}}(x) := \sum_{Q \in \mathcal{E}} |Q|^{-1/s'} \chi_Q(x).$$

We first claim that

$$\|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \lesssim_{N,s} (\#\mathcal{E})^{1/s'}. \tag{5.33}$$

Indeed, fix

$$x \in \bigcup_{Q \in \mathcal{E}} Q.$$

Set

$$\mathcal{E}_x := \{Q \in \mathcal{E} : x \in Q\}.$$

Since $\mathcal{E}$ is finite, there exists a cube $Q_x \in \mathcal{E}_x$ having minimal volume. Since any two cubes in the same dyadic grid which contain the same point are nested, every $Q \in \mathcal{E}_x$ contains Q_x. Thus the cubes in $\mathcal{E}_x$ belong to the chain of dyadic ancestors of Q_x.

Denote by $Q_x^{(k)}$ the k-th dyadic ancestor of Q_x, so that

$$|Q_x^{(k)}| = 2^{kN} |Q_x|, \qquad k \geq 0.$$

It follows that

$$\begin{aligned} G_{\mathcal{E}}(x) &= \sum_{Q \in \mathcal{E}_x} |Q|^{-1/s'} \\ &\leq \sum_{k=0}^{\infty} |Q_x^{(k)}|^{-1/s'} \\ &= |Q_x|^{-1/s'} \sum_{k=0}^{\infty} 2^{-kN/s'} \\ &\lesssim_{N,s} |Q_x|^{-1/s'}. \end{aligned}$$

For each $Q \in \mathcal{E}$, define

$$E_Q := \left\{ x \in \bigcup_{P \in \mathcal{E}} P : Q_x = Q \right\}.$$

The sets $\{E_Q\}_{Q \in \mathcal{E}}$ are pairwise disjoint and satisfy

$$E_Q \subset Q.$$

Therefore,

$$\begin{aligned} \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)}^{s'} &= \int_{\mathbb{R}^N} G_{\mathcal{E}}(x)^{s'}\, dx \\ &\lesssim_{N,s} \sum_{Q \in \mathcal{E}} \int_{E_Q} |Q|^{-1}\, dx \\ &= \sum_{Q \in \mathcal{E}} \frac{|E_Q|}{|Q|} \end{aligned}$$

$$\leq \sum_{Q\in\mathcal{E}} 1 = \#\mathcal{E}.$$

Taking the s'-th root proves (5.33).

We now estimate the sum of $\{b_Q\}_{Q\in\mathcal{E}}$. By Hölder's inequality and (5.33),

$$\begin{aligned}
\sum_{Q\in\mathcal{E}} b_Q &= \sum_{Q\in\mathcal{E}} |Q|^{-1/s'} \int_Q |F(x)|\,dx \\
&= \int_{\mathbb{R}^N} |F(x)| \sum_{Q\in\mathcal{E}} |Q|^{-1/s'} \chi_Q(x)\,dx \\
&= \int_{\mathbb{R}^N} |F(x)| G_{\mathcal{E}}(x)\,dx \\
&\leq \|F\|_{L^s(\mathbb{R}^N)} \|G_{\mathcal{E}}\|_{L^{s'}(\mathbb{R}^N)} \\
&\lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.
\end{aligned}$$

Now fix $t > 0$ and let

$$\mathcal{E} \subset \{Q \in \mathbb{D}^0 : b_Q > t\}$$

be an arbitrary finite subcollection. Since $b_Q > t$ for every $Q \in \mathcal{E}$, we have

$$t\,\#\mathcal{E} < \sum_{Q\in\mathcal{E}} b_Q \lesssim_{N,s} \|F\|_{L^s(\mathbb{R}^N)} (\#\mathcal{E})^{1/s'}.$$

If $\mathcal{E} \neq \emptyset$, dividing by $(\#\mathcal{E})^{1/s'}$ and using

$$1 - \frac{1}{s'} = \frac{1}{s},$$

we obtain

$$(\#\mathcal{E})^{1/s} \lesssim_{N,s} t^{-1} \|F\|_{L^s(\mathbb{R}^N)}.$$

Consequently,

$$\#\mathcal{E} \lesssim_{N,s} t^{-s} \|F\|^s_{L^s(\mathbb{R}^N)}.$$

Since this estimate holds for every finite subcollection

$$\mathcal{E} \subset \{Q \in \mathbb{D}^0 : b_Q > t\},$$

it follows that

$$\#\{Q \in \mathbb{D}^0 : b_Q > t\} \lesssim_{N,s} t^{-s} \|F\|^s_{L^s(\mathbb{R}^N)}.$$

This proves (5.32), and hence also the stated weak ℓ^s estimate. □

Lemma 5.7. For every $I \in \mathbb{D}_\Gamma$, we have

$$\sup_{x\in I} w(x) \leq c_{\kappa,\Phi} \frac{w(I)}{|I|}.$$

Proof. We have

$$w(I) \approx w(c_I, d_I) \approx d_I^N \prod_{v\in R} (|\langle c_I, v\rangle| + d_I)^{k(v)}.$$

On the other hand, if $x \in I$, then $|x - c_I| \leq c_\Phi d_I$ and, therefore,

$$|\langle x, v\rangle| \leq |\langle x - c_I, v\rangle| + |\langle c_I, v\rangle| \leq |x - c_I| + |\langle c_I, v\rangle| \leq c_\Phi d_I + |\langle c_I, v\rangle|.$$

Thus,

$$w(x) \leq \prod_{v \in R} (c_\Phi d_I + |\langle c_I, v \rangle|)^{k(v)} \leq c_{\Phi,\kappa} \prod_{v \in R} (|\langle c_I, v \rangle| + d_I)^{k(v)}.$$

This completes the proof. □

Lemma 5.8. For every $I \in \mathbb{D}_\Gamma$, we have

$$w(I)^{-\frac{1}{r}} \|f\|_{L_r(I, w(x)dx)} \leq c_{\kappa,\Phi} |I|^{-\frac{1}{r}} \|f\|_{L_r(I)}.$$

Proof. The assertion follows immediately from Lemma 5.7. □

Lemma 5.9. We have

$$\|M_f V\|_{\mathcal{L}_{N,\infty}(L_2(\mathbb{R}^N, w(x)dx))} \leq c_{K,\kappa,\Phi} \left\| \left\{ d_I^{1-\frac{N}{r}} \|f\|_{L_r(I)} \right\}_{I \in \mathbb{D}_\Gamma} \right\|_{l_{N,\infty}}.$$

Proof. The assertion follows by combining Lemma 5.5 and Lemma 5.8. □

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